

Normalization in DBMS

What is Normalization?

Definition

Normalization is the process of organizing data in a database to:

- Reduce data redundancy (duplicate data)
- Eliminate data anomalies
- Improve data consistency
- Maintain data integrity
- Make the database easier to manage

Simple Definition

Normalization is a technique used to divide large tables into smaller related tables to reduce duplication and improve data consistency.

Why Do We Need Normalization?

Without normalization, databases may have:

- Duplicate (redundant) data
- Inconsistent information
- Wasted storage
- Difficult updates
- Incorrect records

Example

StudentID	StudentName	Course	Instructor
101	Rahul	Java	John
102	Priya	Java	John
103	Amit	Java	John

Here,

The instructor name **John** is repeated many times.

If John's name changes to **jake**, every row must be updated.

This causes problems.

What is Data Redundancy?

Definition

Data Redundancy means storing the **same information multiple times** in a database.

Example

EmpID	Employee	Department
101	Rahul	HR
102	Priya	HR
103	Amit	HR

The department **HR** is repeated.

This increases storage and creates maintenance problems.

What are Database Anomalies?

Definition

A **Database Anomaly** is an unwanted problem caused by storing redundant data.

These anomalies make data inconsistent and difficult to maintain.

There are **three major anomalies**:

1. Insertion Anomaly
 2. Update Anomaly
 3. Deletion Anomaly
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Insertion Anomaly

Definition

An **Insertion Anomaly** occurs when we **cannot insert new information** without inserting unrelated data.

Example

StudentID	StudentName	Course
101	Rahul	Java
102	Priya	Python

Suppose a new course **SQL** has started, but **no student has enrolled yet**.

Can we insert SQL?

 No.

Because StudentID and StudentName are required.

Problem

Cannot store course information independently.

Solution

Create separate tables.

Student

StudentID	StudentName
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Course

CourseID	CourseName
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Now a course can be added even if no students are enrolled.

Update Anomaly

Definition

An **Update Anomaly** occurs when the **same information exists in multiple rows**, and every row must be updated.

Example

Student	Course	Instructor
Rahul	Java	John
Priya	Java	John
Amit	Java	John

Suppose John changes his name to **jake**.

We must update all rows.

If one row is missed:

```
|Rahul|Java|jake|
|Priya|Java|John|
```

Now the database becomes inconsistent.

Solution

Store instructor details in a separate table.

Deletion Anomaly

Definition

A **Deletion Anomaly** occurs when deleting one record unintentionally deletes important information.

Example

Student	Course
Rahul	Java
Priya	Python

Suppose Rahul leaves the course.

After deleting Rahul,

Student	Course
Priya	Python

Now the information that **Java course existed** is lost.

Solution

Store courses in a separate table.

Deleting students will not remove course information.

Summary of Anomalies

Anomaly	Problem	Example
Insertion	Cannot insert data independently	Cannot add a new course without a student
Update	Need to update the same data multiple times	Instructor name repeated
Deletion	Deleting one record removes important information	Deleting last student removes course information

Goals of Normalization

Normalization aims to:

- Reduce duplicate data
 - Remove anomalies
 - Improve consistency
 - Maintain integrity
 - Make updates easier
 - Save storage space
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Advantages of Normalization

✓ Reduces Redundancy

Duplicate data is minimized.

✓ Eliminates Anomalies

Prevents insertion, update, and deletion problems.

✓ Improves Consistency

Every piece of information is stored only once.

✓ Better Data Integrity

Relationships between tables remain accurate.

✓ Easier Maintenance

Updating one record updates the information everywhere.

✓ Efficient Database Design

Tables become cleaner and more organized.

Disadvantages of Normalization

Although normalization improves database design, it also has some drawbacks:

More Tables

Data is divided into multiple tables.

More Joins

Queries require JOIN operations to retrieve data.

Slightly Slower Read Operations

Complex queries may take more time because multiple tables are accessed.

More Complex Design

Database structure becomes more difficult to understand for beginners.

Note: These disadvantages are usually acceptable because the benefits of consistency and reduced redundancy outweigh them.

Conclusion

- Normalization is a **database design technique** used to organize data efficiently.
- It **reduces redundancy** and **eliminates anomalies**.
- It improves **consistency**, **integrity**, and **maintainability** of the database.
- Normal forms (**1NF**, **2NF**, **3NF**, **BCNF**, **4NF**, and **5NF**) are applied step by step to achieve a well-structured database.